Modelling of Electrodynamic Systems

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Modelling of Electrodynamic Systems

MM8. Oblique Incident Wave

Recall from the previous lecture

How to Learn Field Theory

1. Definition of fields (static fields)
2. Physical laws. (in vector notation)

3. Wave equations. (standard solution)
4. Lossless propagation of plane wave (reflections)

5. Lossy media (absorption)
6. Oblique incident wave

7. Propagation of spherical and cylindrical waves
8. Generation of waves

ETC.

Oblique Incident Wave

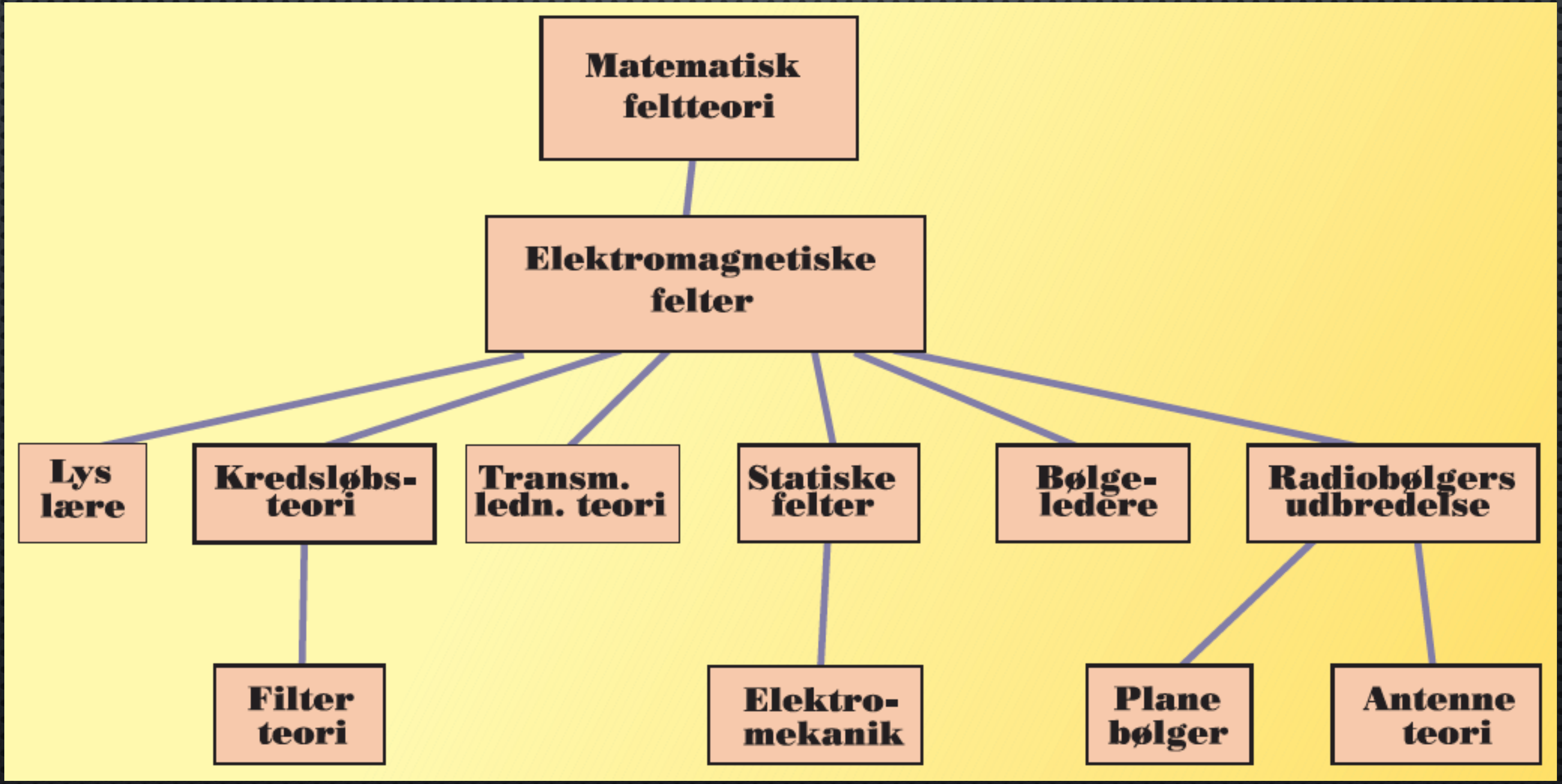
Targets:

1. Read “Grundlæggende Maxwellsk Feltteori” (Page 69-81, 129-151) and “Grundlæggende Transmissionsledningsteori” (Page 93 – 104, 114) (before or after the lecture)

2. Be able to calculate reflected and refracted field for oblique incident waves (lecture)

3. Finish the exercise (after the lecture)

Research Areas Related to Field Theory



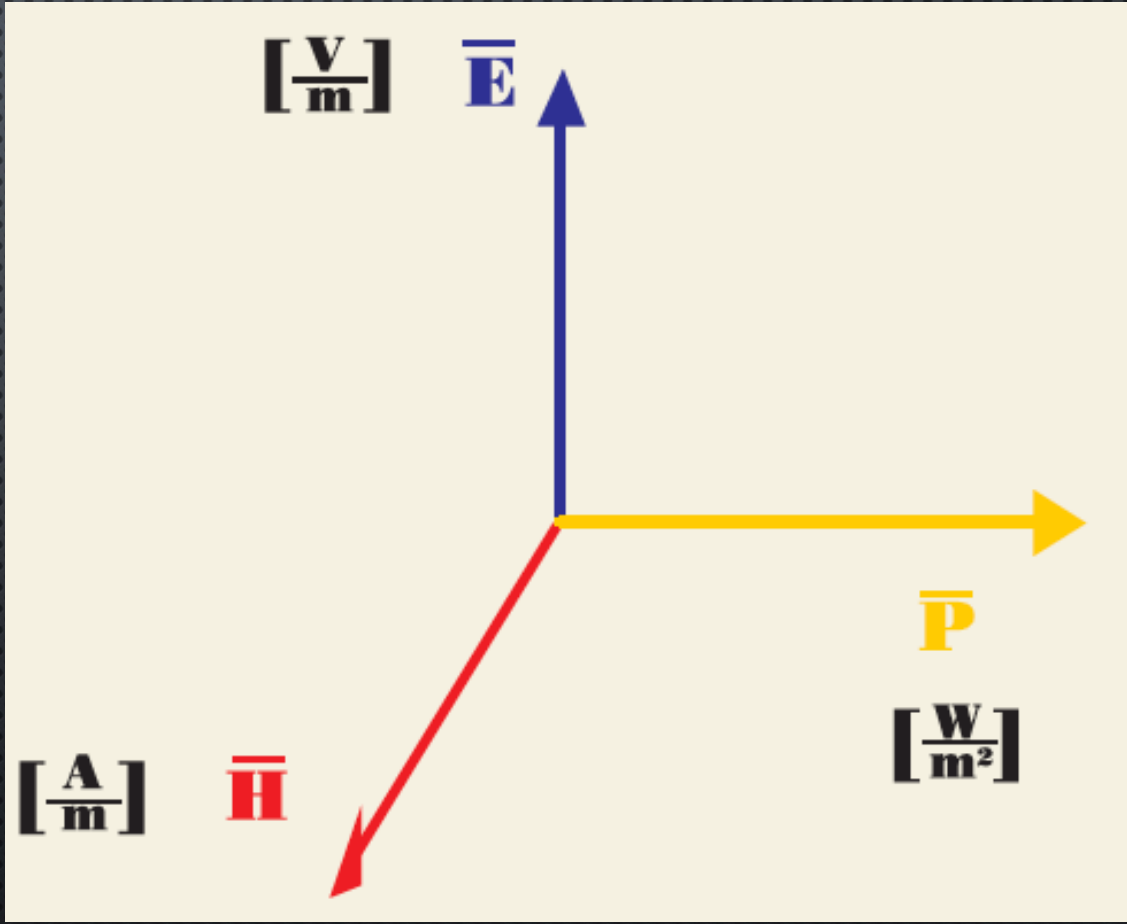
Field Propagation

$$\nabla \times \overline{\mathbf{E}} = -j\omega\mu\overline{\mathbf{H}}$$

$$\nabla \times \overline{\mathbf{H}} = (j\omega\varepsilon + \sigma)\overline{\mathbf{E}}$$

$$\nabla \cdot \overline{\mathbf{B}} = 0$$

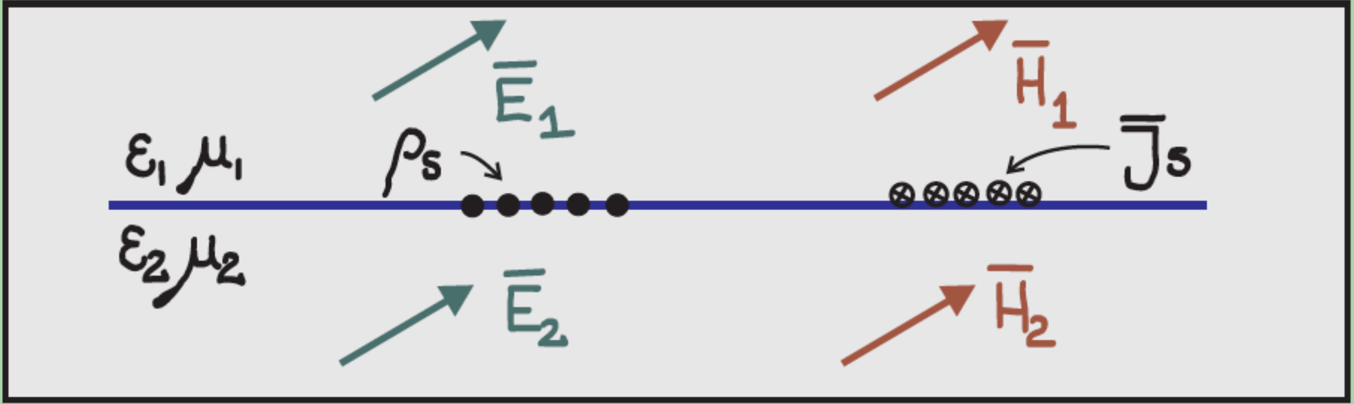
$$\nabla \cdot \overline{\mathbf{D}} = \rho$$



E and H field are orthogonal with each other. Both are also orthogonal to propagation direction. P=E x H

Boundary Condition

General



Electric field

$\nabla \times \bar{\mathbf{E}} = -j\omega\mu\bar{\mathbf{H}}$

$$E_{t1} = E_{t2}$$

$\nabla \cdot \bar{\mathbf{D}} = \rho$

$$D_{n1} = D_{n2} + \rho_s$$

$$\varepsilon_1 E_{n1} = \varepsilon_2 E_{n2} + \rho_s$$

Magnetic field

$$H_{t1} = H_{t2} + J_s$$

$\nabla \times \bar{\mathbf{H}} = (j\omega\varepsilon + \sigma)\bar{\mathbf{E}}$

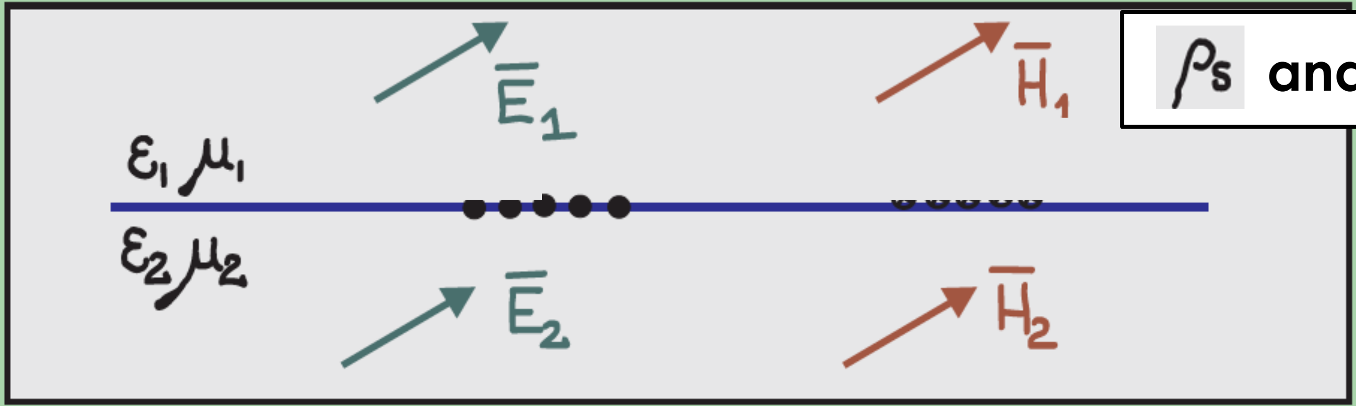
$$B_{n1} = B_{n2}$$

$\nabla \cdot \bar{\mathbf{B}} = 0$

$$\mu_1 H_{n1} = \mu_2 H_{n2}$$

Boundary Condition

Special Case1: Ideal Dielectric



ρ_s and $\bar{J}_s$ are 0

Electric field

$$E_{t1} = E_{t2}$$

$$D_{n1} = D_{n2}$$

$$\varepsilon_1 E_{n1} = \varepsilon_2 E_{n2}$$

Magnetic field

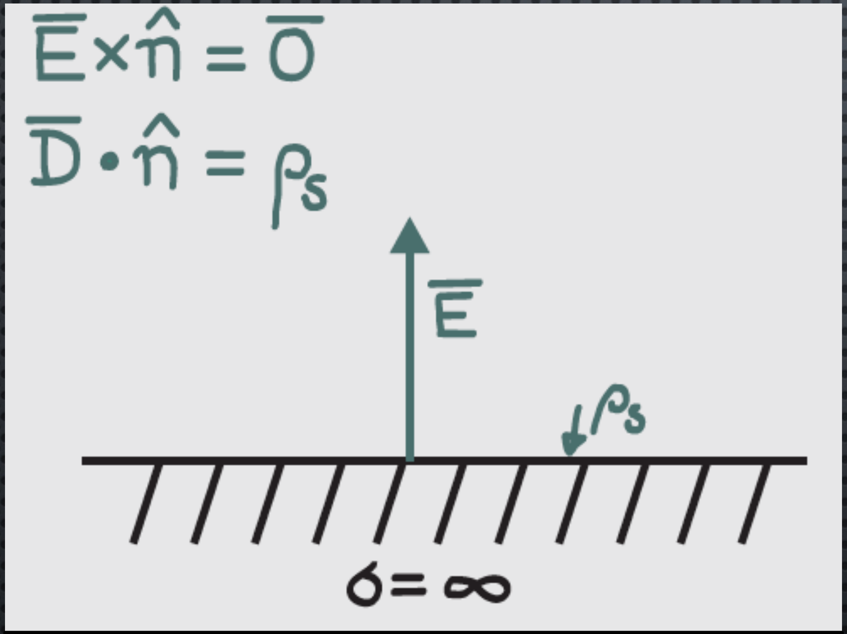
$$H_{t1} = H_{t2}$$

$$B_{n1} = B_{n2}$$

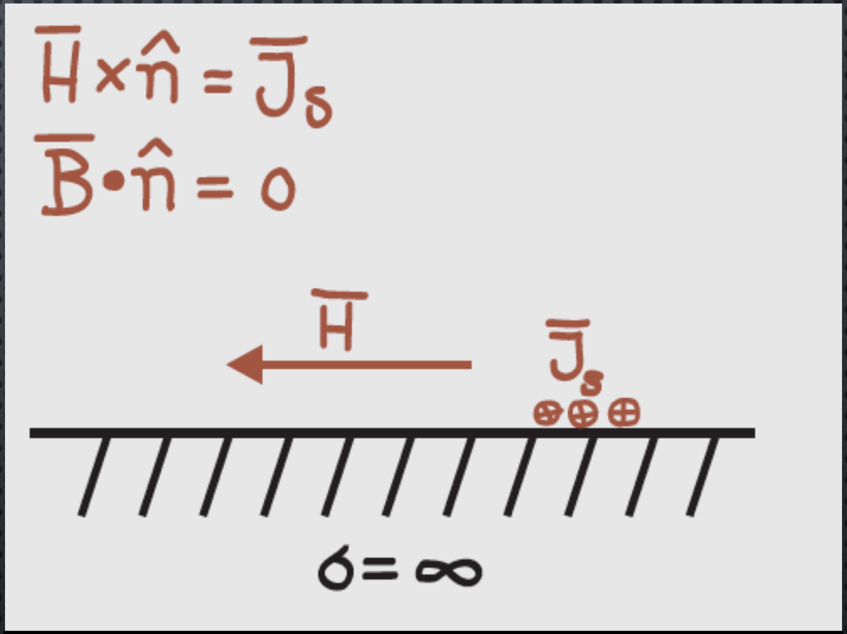
$$\mu_1 H_{n1} = \mu_2 H_{n2}$$

Boundary Condition

Special Case2: Ideal Conductor



E-field is orthogonal to the interface and the surface charges.



H-field is parallel with the interface and orthogonal to the surface currents.

Note!!

The conductor model is a special model that may only be used on special occasions.

Blackboard (1)(2)(3)

Rayleigh Reflection Coefficient

For obliquely incident waves in lossless media, these reflection coefficients are used:

For horizontal polarization

$$K_L = \frac{\frac{1}{\sqrt{\varepsilon_{r2}} \cdot \cos(\Theta_2)} - \frac{1}{\sqrt{\varepsilon_{r1}} \cdot \cos(\Theta_1)}}{\frac{1}{\sqrt{\varepsilon_{r2}} \cdot \cos(\Theta_2)} + \frac{1}{\sqrt{\varepsilon_{r1}} \cdot \cos(\Theta_1)}}$$

For vertical polarization

$$K_L = \frac{\frac{1}{\sqrt{\varepsilon_{r2}}} \cdot \cos(\Theta_2) - \frac{1}{\sqrt{\varepsilon_{r1}}} \cdot \cos(\Theta_1)}{\frac{1}{\sqrt{\varepsilon_{r2}}} \cdot \cos(\Theta_2) + \frac{1}{\sqrt{\varepsilon_{r1}}} \cdot \cos(\Theta_1)}$$

Brewster Angle

Brewster angle is also called polarization angle, which has perfect matching and no reflection.

$$\eta_1 \cdot \cos\Theta_1 = \eta_2 \cdot \cos\Theta_2 \qquad (1)$$

$$\frac{\sin\Theta_1}{\sin\Theta_2} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}} \qquad (2)$$

Fra (2) fås:

$$\sin^2\Theta_2 = \frac{\varepsilon_1}{\varepsilon_2} \cdot \sin^2\Theta_1$$

Fra (1) fås nu:

$$\cos\Theta_1 \cdot \sqrt{\frac{\mu}{\varepsilon_1}} = \cos\Theta_2 \cdot \sqrt{\frac{\mu}{\varepsilon_2}}$$

$$= \sqrt{\frac{\mu}{\varepsilon_2}} \sqrt{1 - \sin^2\Theta_2}$$

$$= \sqrt{\frac{\mu}{\varepsilon_2}} \sqrt{1 - \frac{\varepsilon_1}{\varepsilon_2} \cdot \sin^2\Theta_1}$$

Vi sætter $\Theta_1 = \Theta_B$

$$\cos\Theta_B = \sqrt{\frac{\varepsilon_1}{\varepsilon_2}\left(1 - \frac{\varepsilon_1}{\varepsilon_2}\sin^2\Theta_B\right)}$$

$$\cos^2\Theta_B = 1 - \sin^2\Theta_B = \frac{\varepsilon_1}{\varepsilon_2} - \left(\frac{\varepsilon_1}{\varepsilon_2}\right)^2 \sin^2\Theta_B$$

$$\sin^2\Theta_B = \frac{1 - \frac{\varepsilon_1}{\varepsilon_2}}{1 - \left(\frac{\varepsilon_1}{\varepsilon_2}\right)^2} = \frac{\varepsilon_2}{\varepsilon_1 + \varepsilon_2}$$

$$\sin\Theta_B = \sqrt{\frac{\varepsilon_2}{\varepsilon_1 + \varepsilon_2}}$$

$$\operatorname{tg}\Theta_B = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

Note:
Only for vertical polarization.
Occurs for both $\varepsilon_1 > \varepsilon_2$ **and** $\varepsilon_2 < \varepsilon_1$

Critical Angle

Critical angle gives full reflection.

$\eta_L = 0, \infty$ eller imaginær giver fuldstændig refleksion (Randen af Smithkortet)

$$K_0 = \frac{0-\eta_0}{0+\eta_0} = -1$$

$$K_\infty = \frac{\infty-\eta_0}{\infty+\eta_0} = +1$$

$$K_{jx} = \frac{jx-\eta_0}{jx+\eta_0} = \frac{(jx-\eta_0)^2}{(jx+\eta_0)(jx-\eta_0)} = 1\angle tg^{-1}\left(\frac{2}{\frac{x}{\eta_0}-\frac{\eta_0}{x}}\right)$$

Note:
Occus for both vertical and horizontal polarization.
Only for both $\varepsilon_1 > \varepsilon_2$

$$\eta_{IN} = \eta_2 \cdot \cos\Theta_2 = \eta_2 \cdot \sqrt{1-\sin^2\Theta_2}$$

Endvidere:

$$\frac{\sin\Theta_1}{\sin\Theta_2} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}} \Rightarrow \sin^2\Theta_2 = \frac{\varepsilon_1}{\varepsilon_2}\sin^2\Theta_1$$

Hvorved:

$$\eta_{IN} = \eta_2 \cdot \sqrt{1-\frac{\varepsilon_1}{\varepsilon_2}\sin^2\Theta_1}$$

Bliver imaginær for

$$\frac{\varepsilon_1}{\varepsilon_2}\sin^2\Theta_1 > 1$$

$$\sin\Theta_1 > \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

Dvs. kun løsning for $\varepsilon_1 > \varepsilon_2$